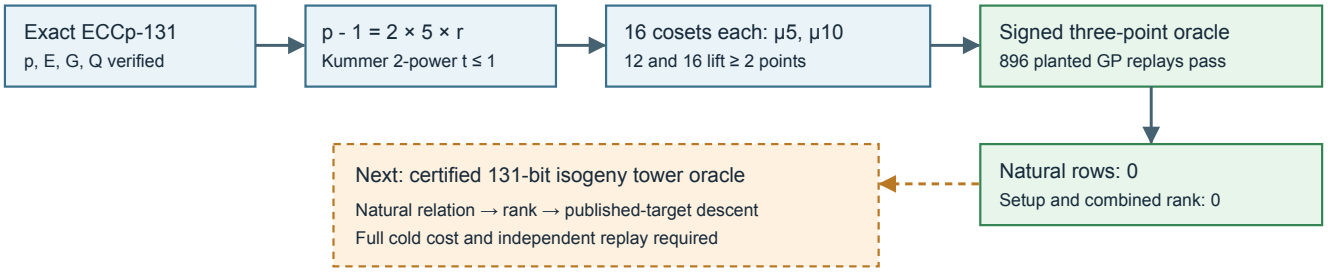
ECCp-131 structured-coset relation screen

Exact-curve bounded observation · 22 October 2026

The published ECCp-131 curve and target were checked at full coordinates. The field has $p - 1 = 2 \cdot 5 \cdot r$ with prime $r = 155003179783434785924857641481313994241$, so a Kummer 2^t-root tower over this field permits only $t \leq 1$. We screened five-root and ten-root multiplicative cosets as separately frozen structured-base controls.



Frozen workload

For each representative $g = 1, \ldots, 16$, lift $g\mu_5$ or $g\mu_{10}$ to the curve. Twelve and sixteen cosets respectively had at least two points. Each was paired with an equal-column small-abscissa base. A complete signed pair table plus subtraction of one signed point tested three-point relations. Seeds 131 and 132 each tested 256 setup and 256 published-target candidates per arm, with eight planted witnesses per arm.

Observations and replay

Base / seed	Completed arms	Planted controls	Natural rows / rank	Cold wall
μ5 / 131	12 coset + 12 control	192 / 192	0 / 0	12.82 s
μ5 / 132	12 coset + 12 control	192 / 192	0 / 0	19.03 s
μ10 / 131	16 coset + 16 control	256 / 256	0 / 0	38.44 s
μ10 / 132	16 coset + 16 control	256 / 256	0 / 0	43.05 s

All 896 planted point identities, both public points, subgroup order, and the factorization were independently checked in PARI/GP. At matched base size, the coset and small-abscissa arms had the same distinct signed pair coverage $2B^2$ for $B = 2, \ldots, 8$. Peak RSS was below 25 MB in each process. The shared host and four cold runs support no speed ranking.

Scope and next gate

A base with at most eight abscissae has no more than $8B^3$ signed triples; its uniform-candidate support on this group is below 2^{-118}. The zero-row observation concerns these cosets and budgets. The exact published-target scalar remains undetermined by this screen. The next candidate is an isogeny-form PKM tower with a certified full-width auxiliary curve and map, followed by one independently replayed natural relation before a rank and target-descent trial. Full cold cost is required.

Raw JSON, commands, timings, replay source, source hashes, and the frozen protocol: research/eccp131_structured_ic_20261022/.